

Power BI Sales Project

Report Documentation & Visual Rationale

SalesProject.pbix • 4 Report Pages • Star Schema Data Model

Project Overview

A four-page end-to-end **Sales Analytics Dashboard** built on a star-schema data model. It gives stakeholders a single pane of glass for monitoring revenue, profitability, geographic reach, product mix, and customer behaviour — all cross-filterable within one report.

Data Model

Table	Role	Key Fields
DimProduct	Product dimension	ProductName, Category
DimGeography	Geography dimension	Country, Region
DimCustomer	Customer dimension	CustomerName, Gender, LoyaltyTier
Dates	Date dimension	Year, Quarter, Month
Measure Table	Central KPI store	Total Sales, Total Profit, YTD/MTD/MTD Sales, YoY Growth %, Total Orders, Sales LY, Avg Unit Price

Pages at a Glance

Pg 1	Sales Overview Executive snapshot of total sales, profit, orders, YoY growth and time trend. KPI Cards · Line Chart · Donut Chart · Gauge
Pg 2	Geographic Sales Analysis Spatial breakdown of sales, MTD, and last-year figures across regions. Map Visual · Country Slicer
Pg 3	Product Analysis Category trends, profit mix, volume metrics, and product-level drillthrough. KPI Visuals · Line Chart · Donut · Cards · Table
Pg 4	Customer Analysis Gender breakdown, loyalty segmentation, orders vs sales, and CRM-level detail. Bar Chart · Scatter Chart · Table · Year Slicer

Page Purpose

Entry point for executives. Communicates overall business health instantly — total revenue, profit, order volume and year-to-date figures — while providing temporal context through a trend line and a performance-vs-last-year gauge.

Key Metrics

Total Sales	Total Profit	Total Orders	YTD Sales	YoY Growth %	Sales LY
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Visuals & Selection Reasoning

Card Visuals (x5) — Total Sales, Total Profit, Total Orders, YTD Sales, YoY Growth %

Cards surface a single KPI with maximum clarity — no axes, no legend, no cognitive load. When an executive opens the report the first question is always 'what are the headline numbers?'; cards answer that instantly. Five separate cards are used (rather than a table) so each metric has equal visual weight and can be read in isolation.

Line Chart — Total Sales over Date

Time-series data is continuous by nature, and a line chart encodes continuity better than any other type. It lets viewers trace momentum, identify seasonality and spot anomalies across the full date range. A bar chart would fragment the timeline and obscure whether the trend is smooth or volatile.

Donut Chart — Total Sales by Category

Part-of-whole composition questions — 'what share of revenue does each category contribute?' — are naturally answered by a donut. With fewer than 6 segments it reads immediately. The hollow centre allows embedding a grand total, adding information density without extra space.

Gauge — Total Sales vs Sales LY (target)

A gauge is the only visual that adds a target-vs-actual dimension to a single value. By anchoring the needle against last year's sales, it tells viewers instantly whether performance is ahead or behind — no mental arithmetic needed. No other visual on this page provides this contextual comparison.

Page Purpose

Answers 'where is revenue coming from?'. By plotting sales on a geographic canvas it makes regional patterns, market concentration and growth opportunities immediately visible. The country slicer enables focused drill-down without leaving the page.

Key Metrics

Total Sales by Region	Sales LY	MTD Sales
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Visuals & Selection Reasoning

Map Visual — Region, Total Sales, Sales LY, MTD Sales

Geography is an inherently spatial dimension — no table or bar chart can communicate 'where' as effectively as a map. Bubble size encodes sales magnitude directly on the physical location, so regional distribution is absorbed in seconds. Tooltips surface MTD and last-year figures without adding visual noise to the canvas. No other chart type can replicate this spatial encoding.

Slicer — Country

A slicer turns the entire page into an interactive drill-down tool. Selecting a country cross-filters the map and any other visuals on the page, giving regional focus without navigating away. It is chosen over the filter pane because it is always visible, one-click, and requires no training.

Page Purpose

Provides a full view of the product portfolio — which categories are growing, which are most profitable, and how individual products rank. It answers both strategic questions (category health) and operational ones (which SKUs to act on).

Key Metrics

Unique Products Sold	Total Quantity	Avg Unit Price	YTD Sales	QTD Sales	MTD Sales	YoY Growth %
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Visuals & Selection Reasoning

KPI Visuals (x3) — YTD Sales, QTD Sales, MTD Sales vs Sales LY

KPI visuals are superior to plain cards for time-bucketed targets because they combine the current value with a directional trend indicator — up/down arrow and % change vs the comparison period. A product manager needs to know not just the YTD number but whether it is running ahead or behind last year; the KPI visual answers both questions in one compact widget.

Line Chart — Total Sales by Year, broken by Category

Multi-series lines are the standard for comparing trend trajectories across groups over time. Using Category as the series colour lets the viewer immediately see which product lines are growing vs declining and whether performance gaps are widening or narrowing — insight that a bar chart would fragment across discrete time buckets.

Donut Chart — Total Profit by Category

Profit mix is a different story from revenue mix — a category can be high-revenue but low-margin. Placing a profit donut alongside the revenue line chart lets the viewer compare both compositions at a glance. The donut is chosen because the question is 'what share of profit' not 'how much profit' — composition questions suit part-of-whole charts.

Card Visuals (x3) — Unique Products Sold, Total Quantity, Avg Unit Price

These three cards provide supply-side context missing from revenue charts. Knowing how many distinct products sold, total volume, and average price helps distinguish revenue growth driven by volume from growth driven by price — a critical strategic distinction for any product team.

Table — ProductName, Total Sales, YTD Sales, YoY Growth %

Charts answer 'what is the shape of the data'; tables answer 'what are the exact numbers'. For product managers who need to identify specific SKUs for action — e.g. the five worst-performing products by YoY Growth % — a sortable table is indispensable. It pairs with the charts: charts show the pattern, the table surfaces the names behind the data.

Page Purpose

Segments and profiles the customer base to answer: who is buying, how often, and how much? It combines demographic breakdown, loyalty segmentation, and individual-level detail to support both marketing strategy and direct CRM follow-up.

Key Metrics

Total Sales by Gender	Total Orders	YTD/QTD/MTD Sales	YoY Growth %	Total Cost	LoyaltyTier
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Visuals & Selection Reasoning

Bar Chart — Gender by YTD Sales, QTD Sales, MTD Sales, Total Sales

Bar charts are the gold standard for comparing discrete categorical groups across multiple measures. Gender is a nominal dimension with two values — grouped bars let the viewer compare Male vs Female across four time buckets simultaneously. A line chart would falsely imply a continuous relationship between gender categories, which is semantically incorrect.

Scatter Chart — Total Orders (X) vs Total Sales (Y), by LoyaltyTier

The scatter chart is the only visual capable of plotting two continuous variables simultaneously while encoding a third dimension through colour (Loyalty Tier). This reveals customer segments organically — e.g. high-order low-spend vs low-order high-spend clusters — which no bar, line or donut chart can surface. It is the highest-information-density visual in the entire report and is uniquely suited to this segmentation question.

Table — CustomerName, Total Orders, Total Sales, YoY Growth %, Total Cost

Customer-level data is granular and action-oriented — a sales team needs to know exactly which accounts to prioritise, not just aggregate trends. A sortable table lets users rank customers by any metric and identify specific names for follow-up. It pairs with the scatter: the chart shows the segment clusters, the table reveals who is in each cluster.

Slicer — Year

A year slicer synchronises all visuals on the page to a selected period, enabling year-over-year cohort comparisons without duplicating the page. Positioned prominently, users naturally set the analytical time context before reading the charts — a simple UX decision that prevents misreading aggregate totals as period-specific figures.